

===== Demo: Secret Matrix Share Size Reduction =====

```
> restart;
> #currentdir("D:\\Dropbox\\Projects\\BAE\\Demo");
> currentdir("D:\\Dropbox\\Projects\\BAE\\Demo");
> read("Secret Sharing v4.mpl");
```

```
> s := 123456789123456789;
s:= 123456789123456789 (1)
```

```
> p := 3;
p:= 3 (2)
```

```
> S := convertSecret(s,p);
S:= 
$$\begin{bmatrix} 0 & 0 & 1 & 2 & 0 & 2 \\ 0 & 1 & 2 & 1 & 1 & 1 \\ 0 & 0 & 2 & 1 & 2 & 2 \\ 2 & 0 & 2 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 & 2 & 1 \\ 2 & 1 & 0 & 1 & 1 & 2 \end{bmatrix}$$
 (3)
```

```
> success := computeCompanionMatrix(S, p);
success:= 
$$\begin{bmatrix} 1 & 2 & 2 & 1 & 1 & 0 \\ 2 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 2 & 1 & 1 & 2 & 1 \\ 1 & 2 & 1 & 1 & 1 & 1 \\ 1 & 1 & 2 & 0 & 2 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 2 & 2 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 2 & 2 & 1 & 0 \end{bmatrix}$$
 (4)
```

$$a^3+3a^2b+3ab^2+b^3 \quad (5)$$